

Launch Countdown

Problem ID: launch-countdown-23-24689126

Mission Control is preparing to launch its new rocket. Before liftoff, N preflight checks must be completed. Some checks can only begin once others are finished. For example, the engines must be fuelled before they can be test fired. Each check has its own independent team, so any checks whose prerequisites are complete can be carried out simultaneously. Check i takes t_i hours to complete. The rocket is ready to launch once every check is complete. Mission Control wants to know which checks are critical. A critical check is a check that if delayed would delay the launch. Find the minimum number of hours until the rocket is ready to launch and the number of critical checks.

Input

The first line contains two integers N and M ($1 \leq N \leq 10^5$, $1 \leq M \leq 10^5$) the number of preflight checks and the number of prerequisite relations. The second line contains N positive integers $t_1, t_2, \dots, t_N$, where t_i is the number of hours needed to complete check i . Checks are numbered from 1 to N . Each of the next M lines contains two integers u and v , meaning check u must be completed before check v can begin. The prerequisite relations contain no cycles.

Output

Output two lines. The first line contains the minimum number of hours until the rocket is ready to launch. The second line contains the number of critical checks.

Hint: A check is critical if it exists in the longest chain of checks. Finding these is very similar to finding the launch time, just inverted.

Sample Input 1

```
5 3
4 3 2 5 2
1 2
2 3
1 5
```

Sample Output 1

```
9
3
```

Sample Input 2

```
6 5
2 3 3 1 6 2
1 2
1 3
2 4
3 4
1 6
```

Sample Output 2

```
6
5
```